CHM-110 | WEEK 10 EXAM CHEAT SHEET

Gases + Thermochemistry • Identify the clue → use the shortest factor-label equation • Quiz questions 1-14

01 Gas stoichiometry, non-STP — 23.8 L

CLUE: Grams of reactant → liters of a different gas; P and T given.

SETUP: V = (6.29 g / 58.12 g mol^{-1}) × (10 mol H_2O / 2 mol C_4H_{10}) × [RT/P]; R = 0.08206, T = 600. K, P = 850/760 atm.

02 Gas stoichiometry at STP — 28.8 L

CLUE: Grams of a reactant → liters of gas; explicitly says STP.

SETUP: 59.2 g Na × (1 mol Na / 22.99 g Na) × (1 mol H_2 / 2 mol Na) × (22.4 L H_2 / 1 mol H_2)

03 Enthalpy sign: exothermic — Exo

CLUE: Negative ΔH given; reaction type requested.

SETUP: ΔH = −73.54 kJ < 0 ⇒ heat RELEASED ⇒ EXOTHERMIC. No calculation required.

04 Calorimetry: find mass — 2.23 kg

CLUE: Energy (J), specific heat and two temperatures; mass requested.

SETUP: m = 5310 J ÷ { [0.130 J/(g·°C)] × (38.3 − 20.0)°C } × (1 kg / 1000 g)

05 Molar enthalpy of solution — +25.7 kJ/mol

CLUE: Given mass (g) + energy (kJ); choices use kJ/mol.

SETUP: ΔH_{mol} = (0.280 kJ / 0.873 g) × (80.052 g / 1 mol) = +25.7 kJ/mol; positive = heat ABSORBED.

06 Concepts and energy diagrams — Conceptual

CLUE: Energy levels, energy placement in equation, warm/cold, or c.

SETUP: ΔH = E(products) − E(reactants). Products higher / heat left = endo; products lower / heat right = exo. c = energy per g per °C.

07 Gas liters → solid grams — 173 g

CLUE: P, V, T of gas product; grams of a different reactant asked.

SETUP: m = [(5.30 atm × 25.0 L) / (0.08206 × 1523.15 K)] × (1 mol $Pb(CO_3)_2$ / 2 mol CO_2) × (327.22 g/mol)

08 Enthalpy sign: exothermic — Exo

CLUE: Negative ΔH given; reaction type requested (same as Q3).

SETUP: ΔH = −54.74 kJ < 0 ⇒ EXOTHERMIC; heat released into surroundings.

09 Ideal Gas Law: pressure — 2.28 atm

CLUE: Given n, V and T; pressure (atm) is unknown.

SETUP: P = nRT/V = [0.545 mol × 0.08206 × (22.0 + 273.15) K] / 5.80 L

10 Thermochemistry: kJ → grams — 357 g

CLUE: Reaction has ΔH (kJ); energy supplied, grams requested.

SETUP: 936 kJ × (2 mol $HC_2H_3O_2$ / 315 kJ) × (60.052 g $HC_2H_3O_2$ / 1 mol $HC_2H_3O_2$)

11 Molecules → liters of gas — 395 L

CLUE: Molecules rather than moles; P and T given; liters asked.

SETUP: V = [(7.32 × 10^{24} molecules) / (6.022 × 10^{23} molecules/mol)] × [0.08206 × 400. K / (768/760 atm)]

12 Ideal Gas Law: grams — 6.46 g

CLUE: P, V, T of ONE gas; ask for mass; no reaction.

SETUP: m = (PV/RT) × M = [3.96 atm × 0.530 L / (0.08206 × 301.15 K)] × 76.02 g/mol

13 STP gas liters → reaction kJ — 114 kJ

CLUE: Gas volume at STP; reaction shows kJ; energy requested.

SETUP: 83.2 L H_2 × (1 mol H_2 / 22.4 L H_2) × (92.0 kJ / 3 mol H_2). NOTE: balanced reaction has $2NH_3$.

14 Calorimetry: find heat q — 5500 cal out

CLUE: Mass + c + initial/final T; energy in cal or J asked.

SETUP: q = mc(Tf − Ti) = 100. g × 1 cal/(g·°C) × (25 − 80)°C = −5500 cal

FAST CONVERSIONS + RECOGNITION RULES

STP (1 atm, 0°C): 1 mol ideal gas ≈ 22.4 L • 760 Torr = 1 atm • 1 L = 1000 mL

T(K) = °C + 273.15 • R = 0.08206 L·atm/(mol·K) • 1 mol = 6.022 × 10^{23} molecules

Different substances in a reaction → use coefficients (mole ratio). No reaction → usually only PV = nRT.

ΔH positive = endothermic / heat absorbed • ΔH negative = exothermic / heat released • q < 0 when cooling